SuppMat_LMEM_function

User Manual — SuppMat_LMEM_function.R

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https://github.com/leonidasjr/SupplementaryMaterials

Supplementary Material

0. About this document

This manual accompanies SuppMat_LMEM_function.R, the R script submitted as supplementary material. It is intended for researchers in linguistics and phonetics who are (un)familiar with R or mixed-effects modelling. The document describes every line of the 85-line script in the order in which it executes, explains the statistical choices made, and provides step-by-step usage instructions.

1. What the script does

`SuppMat_LMEM_function.R` reads the tab-separated dataset `SuppMat_ds_inferential_prosodic_features.txt` produced by `SpeechRhythmExtractor.praat` script and performs, for each of its 34 acoustic parameters, a Linear Mixed-Effects Model (LMEM) of the form:

- **`d_var ~ DIALECT + (1 | SPEAKER)`**
    - where DIALECT is the between-group **fixed effect** (**PB** vs. **SP**) and SPEAKER is the **random intercept**. The function automatically filters out non-significant parameters based on p-value ($p > 0.05$) and reports - for each significant one - the full model summary and the R^2 effect sizes based on Nakagawa and Schielzeth (2013) study. The entire output is saved to `SuppMat_lmem_report.txt` in the same folder as the input dataset.

1.1 The LMEM formula

For each dependent variable d_var, the model is:

```
d_var ~ DIALECT + (1 | SPEAKER)
```

- `DIALECT` (fixed effect): tests whether dialect group membership (PB or SP) significantly predicts `d_var`.
- `(1 | SPEAKER)` (random effect): estimates a random baseline shift per speaker, accounting for the fact that observations from the same speaker are not independent. This structure is the standard approach for inferential phonetics experiments with multiple observations per speaker.

2. Required software and packages

2.1 R installation

R version 4.0 or higher. Download from https://www.r-project.org. RStudio (https://posit.co) is recommended but not required.

2.2 Package installation

Run the following command once in R to install all required packages:

```
install.packages(c('lme4', 'lmerTest', 'emmeans', 'tidyverse', 'MuMIn'))
```

Package	Min. version	Role
`lme4`	≥ 1.1	Core LMEM fitting via lmer()
`lmerTest`	≥ 3.1	Adds Satterthwaite p-values to lmer() summaries
`emmeans`	≥ 1.7	Loaded for post-hoc comparisons (available but not called automatically)
`tidyverse`	≥ 1.3	Data manipulation utilities
`MuMIn`	≥ 1.46	Computes R^2 effect sizes via r.squaredGLMM() (Nakagawa & Schielzeth, 2013)

> ℹ *If you see 'Error: package not found', run the install command above and restart R before trying again.*

3. Input dataset

3.1 Source file

The expected input is `SuppMat_ds_inferential_prosodic_features.txt`, a tab-separated file produced by `SuppMat_SpeechRhythmExtractor.praat`. It contains 460 rows (20 speakers × 23 chunks each) and 39 columns.

3.2 Column structure

Columns 1–5 are metadata; columns 6–39 are the 34 numeric acoustic parameters.

Column	Name	Content
1	`AUDIOFILE`	Source .wav filename
2	`DIALECT`	Dialect group: PB (Paraíba) or SP (São Paulo) — fixed effect
3	`SPEAKER`	Speaker ID: pb01–pb10, sp01–sp10 — random effect (grouping variable)
4	`SEX`	Speaker sex: FEM or MAS
5	`CHUNK`	Utterance chunk label: CH1, CH2 …
6–39	`(34 acoustic params)`	durnorm_Lobanov through artic_rate — all numeric; these become 'd_vars'

> ℹ *The script identifies dependent variables automatically using* **`is.numeric()`**. *Metadata columns (AUDIOFILE, DIALECT, SPEAKER, SEX, CHUNK) are character/factor and are excluded from d_vars automatically.*

4. Step-by-step usage

1. Ensure all five packages are installed (Section 2.2).
2. Place SuppMat_LMEM_function.R and SuppMat_ds_inferential_prosodic_features.txt in an accessible folder.

3. Open RStudio (or R console).
4. Open the file: File → Open File… → select `SuppMat_LMEM_function.R`.
5. Run the script: Source button (top-right of the script editor) or press Ctrl+Shift+Enter (Windows/Linux) / Cmd+Shift+Enter (macOS). Alternatively, run line by line with Ctrl+Enter / Cmd+Enter.
    - QUICK TIP: Select all lines at once with Ctrl+A, and run with Ctrl+Enter (Windows).
6. When the file-chooser dialog appears (Step 1, line 51), navigate to and select `SuppMat_ds_inferential_prosodic_features.txt`.
7. Wait for the analysis to be completed. Progress is shown in the R Console.
8. Read the Console output to see which parameters were significant.
9. Find `SuppMat_lmem_report.txt` in the same folder as the input dataset.

> ℹ *The file-chooser dialog (file.choose()) may not appear in some non-interactive R environments (e.g., Rscript from the terminal). In that case, set data_file manually at the beginning of the script: data_file <- '/full/path/to/SuppMat_ds_inferential_prosodic_features.txt'*

5. Output file — SuppMat_lmem_report.txt

The report is a plain-text file saved in the same folder as the input dataset. It contains one block per significant parameter in the order they appear in the dataset. Each block has this structure:

```
=========== durnorm_Lobanov ===========
Linear mixed model fit by REML. t-tests use Satterthwaite's method
...
Fixed effects:
            Estimate Std. Error    df  t value  Pr(>|t|)
(Intercept)  -0.123     0.045   18.2  -2.733   0.01382 *
DIALECTSP     0.287     0.063   18.2   4.556   0.00024 ***
...
Effect size (Nakagawa & Schielzeth, 2013):
Marginal R² (fixed effects): 0.182
Conditional R² (fixed + random effects): 0.641
```

Figure 1. Example output block for a significant parameter. The separator line contains the variable name. Below it: the lmerTest fixed-effects table (Estimate, SE, df, t, p) and random-effects variance; then the Nakagawa & Schielzeth R^2 values.

> ℹ *Parameters with p > 0.05 on the DIALECT fixed effect are silently skipped. Only the fixed-effect p-values for non-intercept terms (p_values[-1]) are tested.*

5.1 Reading the fixed-effects table

Each significant block contains a fixed-effects table with the following columns:

Column	Meaning
`Estimate`	`Regression coefficient — mean difference between SP and PB on this parameter (in the parameter's original units)`
`Std. Error`	`Standard error of the estimate`
`df`	`Satterthwaite degrees of freedom (from lmerTest) - not necessarily an integer`

Column	Meaning
t value	t-statistic = Estimate / Std. Error
Pr(>|t|)	Two-tailed p-value. Values ≤ 0.05 are flagged as significant by the function

5.2 Reading the R^2 values

Below each model summary, two R^2 values are printed:

- Marginal R^2 (R^2m): proportion of total variance explained by DIALECT alone.
- Conditional R^2 (R^2c): proportion of total variance explained by DIALECT + SPEAKER together.

A large gap between R^2m and R^2c indicates substantial between-speaker variability - the main reason for using a random-effects model rather than a simple t-test.

The table below summarizes all fields that appear in each report block:

Field	Description
REML criterion	Model fit statistic (Restricted Maximum Likelihood). Lower = better fit within a fixed-structure model
AIC / BIC	Akaike and Bayesian Information Criteria — used for model comparison
Random effects	SPEAKER intercept variance: variance of speaker baseline differences. Residual: unexplained within-speaker variation
Fixed effects	Intercept = grand mean (reference level). DIALECTSP = mean difference of SP minus PB reference level
Pr(>|t|)	P-value from Satterthwaite approximation (lmerTest). < 0.05 = significant DIALECT effect
Marginal R^2	Variance explained by DIALECT alone (fixed effect)
Conditional R^2	Variance explained by DIALECT + SPEAKER together

6. Line-by-line code walkthrough

The blocks below document every section of SuppMat_LMEM_function.R. Line ranges refer to the actual script (85 lines total).

Lines 1–5 — Library loading

Lines 1–5 load five R packages before any analysis runs. Each library() call must succeed; if a package is missing, R will stop with an error at this point (see Troubleshooting). The packages are: lme4 for model fitting, lmerTest to add Satterthwaite-corrected p-values to the lmer() summary, emmeans for estimated marginal means (available for post-hoc use), tidyverse for data utilities, and MuMIn for the r.squaredGLMM() function used to compute R^2 effect sizes.

Lines 7–46 — Function significant_mixed_models() — definition

Lines 7–46 define the central function. It takes five arguments:

- d_vars: a character vector of dependent variable names (all numeric columns).
- fixed_effect: a single string naming the between-group factor ("DIALECT").

- random_effect: a single string naming the grouping variable for the random intercept ("SPEAKER").
- data: the data frame.
- alpha: the significance threshold (default 0.05).

The function does NOT return a value; all output is printed to the console (and captured by sink() in Step 5).

Lines 8 — results <- list()

Line 8 initializes an empty list object. Although results is never populated in the current version (the function only prints), the slot is reserved for future extension - e.g., returning model objects to the calling environment.

Lines 10–11 — for loop over d_vars — formula construction

Line 10 begins the loop over every dependent variable. Line 11 constructs the LMEM formula dynamically using as.formula(paste(...)). For example, if d_var = 'durnorm_Lobanov', fixed_effect = 'DIALECT', and random_effect = 'SPEAKER', the resulting formula is:

- `durnorm_Lobanov ~ DIALECT + (1 | SPEAKER)`

This is a random-intercept model: DIALECT is the fixed effect and SPEAKER provides a random baseline shift per speaker, controlling for between-speaker variability.

Lines 13–15 — Random effect validation guard

Lines 13–15 check that the column named in random_effect actually exists in the data frame before attempting to fit any model. If the column is missing, the function stops immediately with an informative error message, preventing a harder-to-diagnose error from lmer().

Lines 17 — Model fitting with try()

Line 17 calls lmer(formula, data = data) inside try(..., silent = TRUE). The try() wrapper catches any R error (e.g., convergence failure, rank-deficient model) and returns an object of class 'try-error' rather than crashing the loop.

Lines 19–22 — Error handling — skip on fitting failure

Lines 19–22 check whether the fitted object is a 'try-error'. If so, a message is printed to identify which variable failed, and next skips to the following iteration of the for loop. This means one failing variable does not abort the analysis of the remaining 33 parameters.

Lines 24–29 — Extract fixed-effect coefficients and p-values

Line 24 extracts the coefficients table from summary(model)$coefficients. This table contains columns Estimate, Std. Error, df, t value, and Pr(>|t|). Line 26 checks that the Pr(>|t|) column is present — it requires lmerTest to be loaded; without it, lmer() returns a summary without p-values and the function skips the variable (lines 27–29).

Lines 31–33 — Significance filter — the selection criterion

Line 31 extracts the full vector of p-values. Line 33 applies the selection criterion:

- length(p_values) > 1 ensures there is at least one non-intercept coefficient (i.e., the fixed effect has at least one level contrast).
- p_values[-1] removes the intercept p-value (index 1) and tests only the fixed-effect slope(s).
- any(...< alpha, na.rm = TRUE) passes if at least one non-intercept coefficient is significant.

Only parameters satisfying this criterion are reported.

Lines 34–35 — Print full model summary

Line 34 prints a visual separator and the dependent variable name. Line 35 calls print(summary(model)), which produces the full lmerTest summary including: fixed-effects table (Estimate, SE, df, t, p); random-effects variance (SPEAKER intercept variance + residual variance); model fit statistics (REML criterion, AIC, BIC, log-likelihood); and number of observations.

Lines 37–43 — R^2 effect size — Nakagawa & Schielzeth (2013)

Line 37 calls MuMIn::r.squaredGLMM(model) inside another try() wrapper. This function returns a two-element vector:

• r2[1] = Marginal R^2 (R^2m): proportion of variance explained by the fixed effects alone (DIALECT).

• r2[2] = Conditional R^2 (R^2c): proportion of variance explained by both fixed and random effects (DIALECT + SPEAKER).

The difference ($R^2c - R^2m$) reflects how much of the total variance is attributable to between-speaker variability. Lines 40–42 print both values rounded to 3 decimal places with their labels.

Lines 44–46 — End of function body

Line 44 closes the if block (significance test). Line 45 closes the for loop. Line 46 closes the function definition. No return value is produced.

Lines 51 — Step 1 — file.choose(): interactive file selection

Line 51 calls file.choose(), which opens a system file-chooser dialog. The user navigates to and selects the dataset file. The full file path (including folder, filename, and extension) is stored in data_file. This is the only interactive step in the script; all subsequent operations are fully automatic.

Lines 56 — Step 2 — read.delim(): load dataset

Line 56 reads the selected file as a tab-separated table (read.delim() defaults to sep = "\t"). stringsAsFactors = TRUE converts character columns (AUDIOFILE, DIALECT, SPEAKER, SEX, CHUNK) to R factors automatically. Numeric columns (the 34 acoustic parameters) are read as numeric.

Lines 61–65 — Step 3 — str() and explicit factor conversion

Line 61 calls str(ds_CadLin), which prints the structure of the data frame to the console: number of observations and variables, column names, types, and a preview of values. This is a sanity check — inspect it to confirm the dataset loaded correctly.

Lines 64–65 explicitly re-convert DIALECT and SPEAKER to factors. This is recommended because lmer() requires grouping variables to be factors, and re-converting ensures consistent level ordering regardless of how stringsAsFactors handled them.

Lines 70–72 — Step 4 — define analysis variables

Line 70 uses sapply(ds_CadLin, is.numeric) to identify all numeric columns in the data frame and stores their names in d_vars. Given the dataset structure, this yields the 34 acoustic parameters (columns 6–39) and excludes the metadata columns (AUDIOFILE, DIALECT, SPEAKER, SEX, CHUNK).

Line 71 sets fixed_effect = "DIALECT" — the between-group categorical predictor.

Line 72 sets random_effect = "SPEAKER" — the random-intercept grouping variable.

Lines 77–81 — Step 5 — sink(): capture output and save report

Line 77 constructs the output path: same folder as the input dataset (dirname(data_file)) with filename SuppMat_lmem_report.txt.

Line 79: sink(output_file, split = TRUE) redirects all console output to the file AND simultaneously displays it in the R console (split = TRUE). Everything printed by the function — headers, model summaries, R^2 values — goes into this file.

Line 80 calls the function with the four defined arguments.

Line 81: sink() with no arguments closes the redirection and restores normal console output.

Lines 83–85 — Completion messages

Lines 83–85 print three messages to the R console (not to the file, since sink() has been closed):

- 'Mixed-Effects Modeling ended.' — signals completion.
- A separator line.
- 'Output saved in: [full path]' — confirms where the report was saved.

7. Troubleshooting

Symptom	Cause and solution
`Error: package 'lme4' not found`	`Run: install.packages(c('lme4','lmerTest','emmeans','tidyverse','MuMIn'))`
`No p-values returned for [variable]`	`lmerTest is not loaded or its Satterthwaite method is unavailable. Ensure library(lmerTest) runs without error.`
`Error fitting model for [variable]`	`The model could not converge (e.g. singular fit, all-zero column, or insufficient variance). The function skips the variable and continues.`

Symptom	Cause and solution
Report file is empty	No parameter had a significant fixed effect (p < 0.05). Check alpha value and dataset content.
file.choose() dialog does not appear	On some systems, RStudio or Rscript may suppress the dialog. Set data_file manually: data_file <- '/path/to/SuppMat_ds_inferential_prosodic_features.txt'
Encoding error reading the file	Ensure the .txt file was saved with UTF-8 or system-default encoding. Try: read.delim(data_file, fileEncoding = 'UTF-8')
Output file not saved	Check write permissions in the folder containing the dataset. Run from a folder where you have write access.

8. Citation

When using this script in publications, please cite:

> *BATES, D.; MÄCHLER, M.; BOLKER, B.; WALKER, S. Fitting linear mixed-effects models using lme4. Journal of Statistical Software, v. 67, n. 1, p. 1–48, 2015. DOI: https://doi.org/10.18637/jss.v067.i01.*
>
> *NAKAGAWA, S.; SCHIELZETH, H. A general and simple method for obtaining R² from generalized linear mixed-effects models. Methods in Ecology and Evolution, v. 4, n. 2, p. 133–142, 2013. DOI: https://doi.org/10.1111/j.2041-210x.2012.00261.x.*

References (ABNT)

BATES, D.; MÄCHLER, M.; BOLKER, B.; WALKER, S. Fitting linear mixed-effects models using lme4. Journal of Statistical Software, v. 67, n. 1, p. 1–48, 2015. DOI: https://doi.org/10.18637/jss.v067.i01.

KUZNETSOVA, A.; BROCKHOFF, P. B.; CHRISTENSEN, R. H. B. lmerTest package: Tests in linear mixed effects models. Journal of Statistical Software, v. 82, n. 13, p. 1–26, 2017. DOI: https://doi.org/10.18637/jss.v082.i13.

NAKAGAWA, S.; SCHIELZETH, H. A general and simple method for obtaining R² from generalized linear mixed-effects models. Methods in Ecology and Evolution, v. 4, n. 2, p. 133–142, 2013. DOI: https://doi.org/10.1111/j.2041-210x.2012.00261.x.

R CORE TEAM. R: A language and environment for statistical computing. Vienna: R Foundation for Statistical Computing, 2024. Disponível em / Available at: https://www.r-project.org.

SILVA JR., L.; SILVA, J.; MEER, P. Prosodic aspects of Brazilian L2 English: A comparison of duration-based rhythm and F0 measures with American English, Indian English, and Brazilian Portuguese. In: Proceedings of. Speech Prosody 2024, p. 101-105, 2024. DOI: 10.21437/SpeechProsody.2024-21.